Brighter Blue

Core Testing>Accelerators & ATA Basics

Instructor Guide

IBM

Brighter Blue

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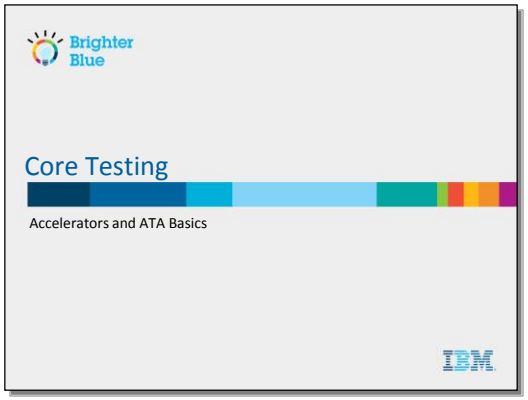
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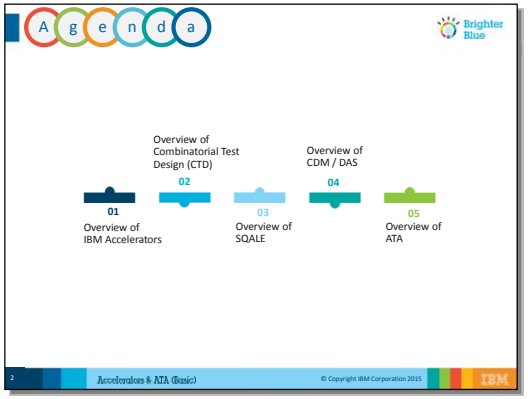
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IBM

Module 01: IBM Accelerators Overview

Slide Content	Instructor Guide	Use this column for your notes
Slide 1 		

Slide Content	Instructor Guide	Use this column for your notes
Slide 2  **At the end of this course, you should be able to:** ▪ Describe IBM Accelerators, such as CTD, SQALE, CDM, and ATA ▪ Identify the use of test accelerators on test life cycle. ▪ Define the basic concept of Modeling, specially CTD ▪ Define the basic concept of Document Analytics	**Purpose:** To cover the training program agenda for Day 1 **Approximate Duration:** 10 minutes **What to cover:** • Distribute the Participant guides and Program attendance sheets • Go over the agenda for this session. Briefly describe the subject areas that will be covered and the activities that will be performed (exercises, case studies). Address the expectations that the participants listed in their introductions. Include information about breaks, lunch, and so on while describing the agenda. • Inform the participants about the evaluation criteria and end of course assessments.	

Slide Content	Instructor Guide	Use this column for your notes
▪ Identify the benefits and barriers of test automation ▪ Describe the usage of ATA	• Inform the participants that you expect highly interactive sessions and would welcome any questions from them. • Ask that all participants turn off their phones or put them in silent mode. Inform them that if they are found to be using their phones or any mobile devices during the sessions, they would need to hand it over to you and collect it from you at the end of the day. • Inform the participants of the location of exits (in case of fires or any other emergencies). • Go over the format of the participant guide. Encourage them to take notes as the course using the Notes pages provided in the guide.	

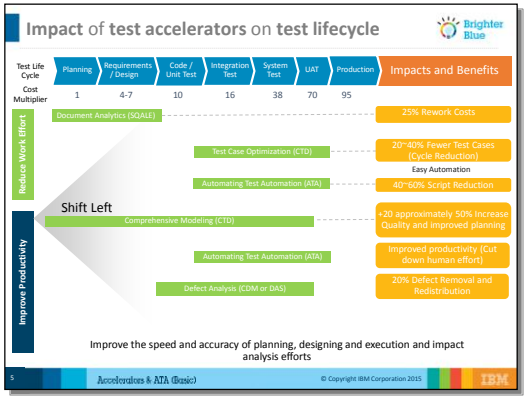
Slide Content	Instructor Guide	Use this column for your notes
Slide 3  At the end of this module, you should be able to: ▪ Recognize the features of IBM Total Test Quality (TTQ) approach ▪ Differentiate between the Traditional test approach and next generation testing	**Purpose:** To cover the objectives of this module **Approximate Duration:** 2 minutes **What to cover:** ▪ Tell the participants what would be covered in this module and what they would be expected to be able to do after they complete this module. ▪ At the end of this module, participants should be able to: ○ Recognize the features of IBM Total Test Quality (TTQ) approach ○ Differentiate between the Traditional test approach and next generation testing	

Slide Content	Instructor Guide	Use this column for your notes
Slide 4  	**Activity title:** What are IBM Accelerators: Discussion **Purpose:** To introduce the concept of Accelerators through a discussion activity **Approximate Duration:** 20 minutes **Additional Material or Pre-activity prep tasks**: ▪ Easel sheets / paper ▪ Pens **Preparation for the Activity:** ▪ Divide the class into three groups. Ask them to say A, B, and C aloud and raise their hands. In this way the whole class gets divided into three groups. ▪ Assign each question to a group as follows	

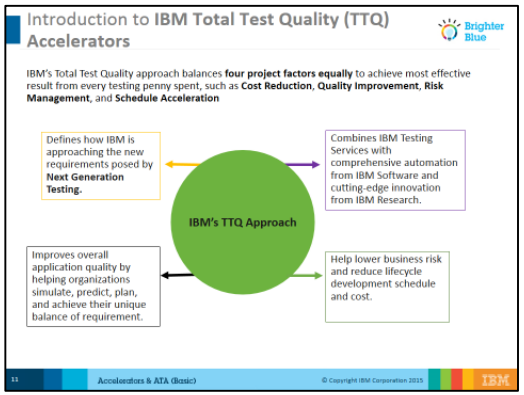
Slide Content	Instructor Guide	Use this column for your notes
	○ Group A: What has been the traditional mode of testing? ○ Group B: Why use Accelerators ○ Group C: What are the Next Generation Accelerators? ***Note**: Depending upon the class size, you can keep this grouping for the later Activities OR you can choose to regroup later.* **tions for the participants:** ▪ Tell them that they need to discuss and come up with the answer to the question based on what they have learned about testing so far. (They would have completed the Basic Testing Course before attending this session).	

Slide Content	Instructor Guide	Use this column for your notes
	▪ Post discussion within the group, any one person needs to share the answer with the rest of the class. ▪ Allow them 5 minutes to discuss and get the answer ready and written on the easel sheets / paper. ▪ Allow each group 2 minutes to present their answers. **Debrief:** ▪ Explain in brief the answer to each question. ▪ Add on to their answers if they answer correctly. Else, explain where they went wrong before sharing the correct answer. ▪ Lastly, point out that the next slide will show them how test accelerators positively impact the testing lifecycle.	

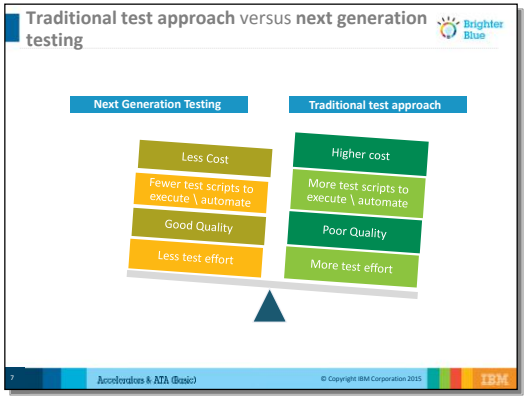
Core Testing>Accelerators & ATA Basics

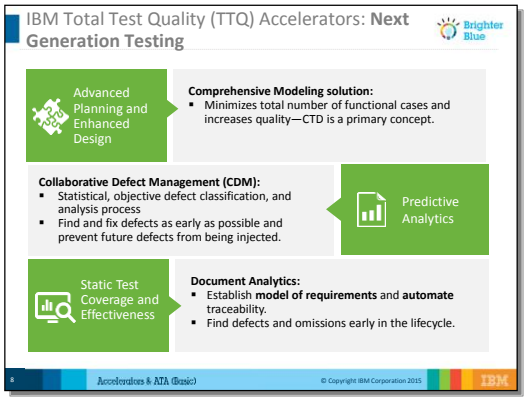
Slide Content	Instructor Guide	Use this column for your notes
Slide 5  **Note** that this diagram is a depiction of high level requirements and the design figures are derived from Boehm and Capers-Jones. The Detailed Requirements Review and Detailed Design Review multipliers are from IBM Research and were developed subsequent to when the earlier models were published, whereas savings are based on solution benchmarks.	**Purpose:** To introduce the concept of Accelerators **Approximate Duration:** 2 minutes **What to cover:** ▪ Make sure that slide is a natural progression from the last slide on Discussion Activity. ▪ Explain the need of accelerators to the participants. ▪ Tell them about the impact and benefits of accelerators in the test lifecycle. ▪ Emphasize on the Cost Multiplier for each phase. ▪ Cover the points in the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

Slide Content	Instructor Guide	Use this column for your notes
Earlier, we had either manual or automated testing but, now there are different levers of testing in the software development lifecycle. An appropriate testing shifts the defect finding activities to the left of SDLC thus reducing total cost of development dramatically. It costs more to fix a defect that is identified at a later stage of project lifecycle than the one which is identified early. However, early testing offers several challenges. IBM understands these challenges and offers several accelerators that are aligned to different phases of testing. These accelerators ensure that we test early (shift left) and test less that prevents critical defects. Some **examples of testing accelerators** are: ▪ An accelerator that minimizes the number of test cases. ▪ Adaptive Test Automation **(ATA) –** a test automation accelerator that automatically converts manual test steps in executable steps in natural language.		

Slide Content	Instructor Guide	Use this column for your notes
Slide 6  IBM's Total Test Quality approach balances **four project factors equally** to achieve most effective result from every testing penny spent, such as **Cost Reduction**, **Quality Improvement**, **Risk Management**, and **Schedule Acceleration**. **IBM's TTQ approach:** ▪ Defines how IBM is approaching the new requirements posed by **Next Generation Testing.**	**Purpose:** To introduce the concept of IBM Total Test Quality (TTQ) Accelerators **Approximate Duration:** 2 minutes **What to cover:** ▪ First tell participants about the following four project factors related to Testing: ▪ Cost Reduction ▪ Quality Improvement ▪ Risk Management ▪ Schedule Acceleration ▪ Now explain how IBM's Total Test Quality (TTQ) approach balances the above **four project factors equally** to achieve most effective result from every testing penny spent ▪ Explain each point listed for the IBM's TTQ approach in the notes section of the slide (also	

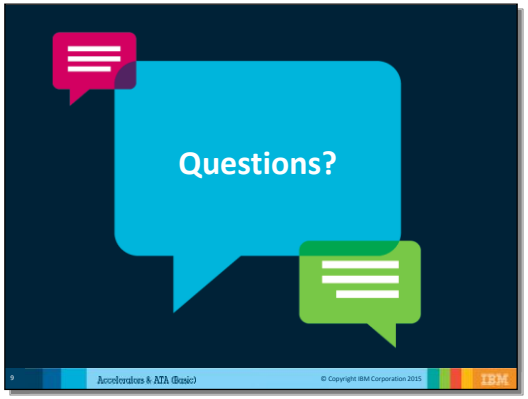
Slide Content	Instructor Guide	Use this column for your notes
▪ Combines IBM Testing Services with comprehensive automation from IBM Software and cutting-edge innovation from IBM Research. ▪ Help lower business risk and reduce lifecycle development schedule and cost. ▪ Improves overall application quality by helping organizations simulate, predict, plan, and achieve their unique balance of requirement.	in the Slide Content Column of this document).	

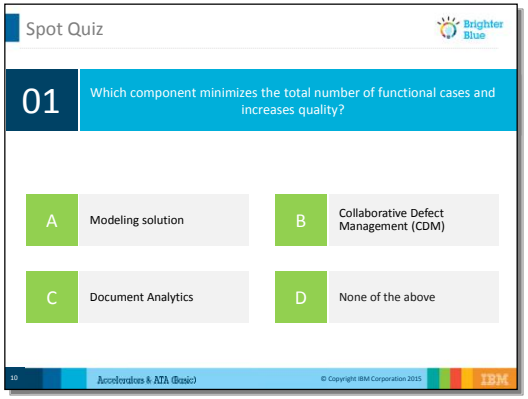
Slide Content	Instructor Guide	Use this column for your notes
Slide 7  In a traditional approach, there are more number of test cases, add more testers, and more test scripts needs to be automated. This takes more time, effort, adds to cost and has a low ROI. The next generation testing or intelligent testing approach—reduces number of test cases and reduce number of testers. This results faster go to market, less cost, and high ROI. In addition, this type of	**Purpose:** To differentiate between traditional and next generation testing **Approximate Duration:** 2 minutes **What to cover:** ▪ Start by writing down the four parameters that differentiate next generation testing from traditional testing: ○ Cost ○ Volume of Test scripts ○ Quality ○ Effort ▪ Point out the differences against each of the above the parameters.	

Slide Content	Instructor Guide	Use this column for your notes
Testing has fewer number of test cases to be automated.		
Slide 8  Next generation testing has been made possible through the collaboration with IBM research and use of innovative test accelerators, such as and Adaptive Test Automation (ATA). **CTD** is an accelerator that minimizes the number of test cases and **ATA** is a test automation accelerator	**Purpose:** To define next generation testing in detail **Approximate Duration:** 2 - 5 minutes **What to cover:** ▪ Introduce the IBM Total Test Quality (TTQ) Accelerators by explaining how Next generation testing has been made possible through collaboration with IBM research and use of innovative test accelerators, such as Combinatorial Test Design (CTD) and Adaptive Test Automation (ATA). ▪ Make sure you cover the points in the notes section of the slide	

Slide Content	Instructor Guide	Use this column for your notes
that automatically converts manual test steps to executable steps in natural language. Software development and testing organization need assistance as they are increasingly involved in critical projects with tight deadlines and low budget. And, the situation goes worse when there is a very large project. So, sometimes, research has seen that the cost of testing takes more than 60 or 70 % time of entire SDLC. Even after that, people feel that the problem is due to defect leakages. Besides that, current testing activities are ad-hoc and it depends upon the personal experience to explore systems, time, and cost. **Features of TTQ** **Comprehensive modelling solution:** Unlike development, system testing is not model or module-based or components-based activities, rather it is flat. So, when we are moving towards next generation testing from traditional-based approach, exhaustive testing in the real world is not enough. Besides, it takes billion and trillion of test cases. Hence, we are	(also in the Slide Content Column of this document). ▪ Explain each of the following TTQ features (given in each section of the diagram on slide) and emphasize the benefit of each (given in the green callouts for each section): ○ Comprehensive modelling solution ○ Collaborative Defect Management (CDM) ○ Document Analytics	

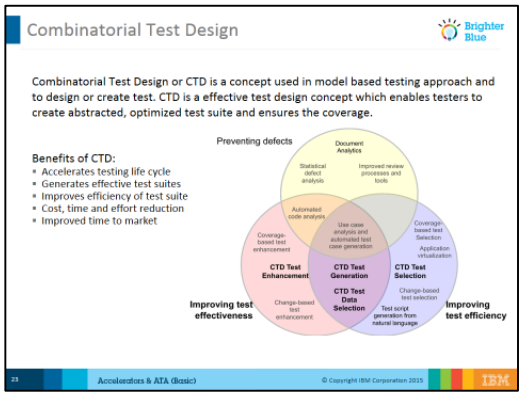
Slide Content	Instructor Guide	Use this column for your notes
trying to bring more modularity-module and model based testing. These are more components driven, function point and test-point driven in terms of models. Modeling solution helps minimize total number of functional cases and can increase quality of the **Collaborative Defect Management (CDM):** This helps in the statistical, objective defect classification, and analysis process. It also helps to find and fix defects as early as possible and prevent future defects from being injected. **Document Analytics**: This helps to establish model of requirements, automate traceability and to find defects and omissions early in the lifecycle, to minimize the cost to correct, the impact on schedule, and the amount of rework.		

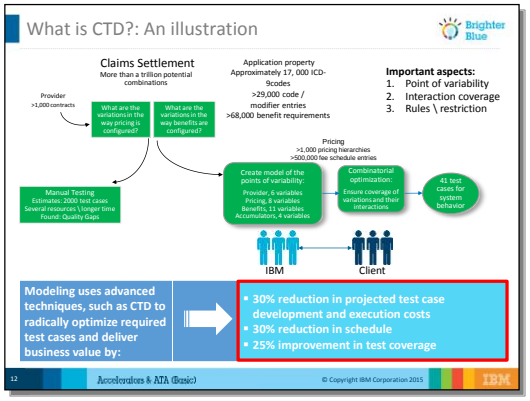
Slide Content	Instructor Guide	Use this column for your notes
Slide 9  	**Approximate Duration:** 2 – 5 minutes **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items ▪ Use this activity to recap the key takeaways from this module	

Slide Content	Instructor Guide	Use this column for your notes
Slide 10  	**Purpose:** To check knowledge of the participants for the topic covered. **Approximate Duration:** 2 minutes **What to cover:** ▪ Ask the participants the question on the slide. ▪ Treat the question as a poll - ask them to raise their hands based on their answer. ▪ The correct answer is **A Modeling solution**	

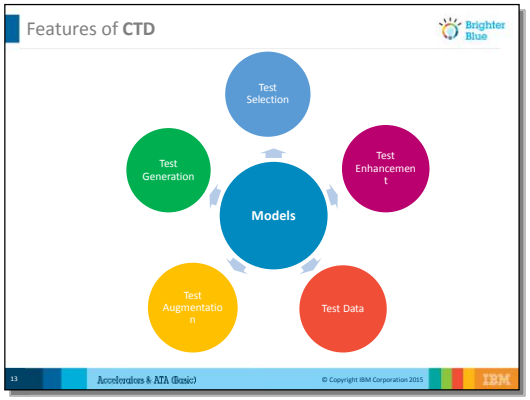
Module 02: Overview of Combinatorial Test Design (CTD)

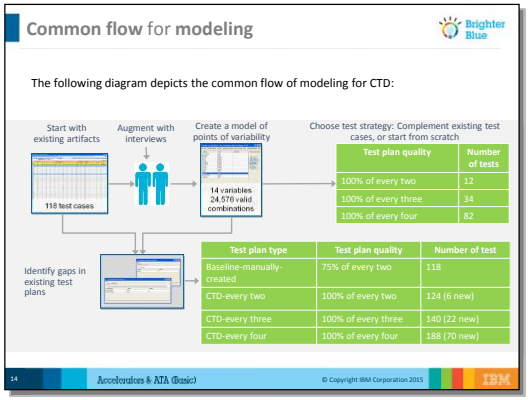
Slide Content	Instructor Guide	Use this column for your notes
Slide 11  At the end of this module, you should be able to: ▪ Define CTD ▪ Explain the common flow for modeling ▪ List the different types of models	**Purpose:** To go over the objectives of this module **Approximate Duration:** 5 minutes depending on length of program **What to cover:** ▪ Provide the participant with the list of topics covered in this module, which includes: ○ Define CTD ○ Explain the common flow for modeling ○ List the different types of models	

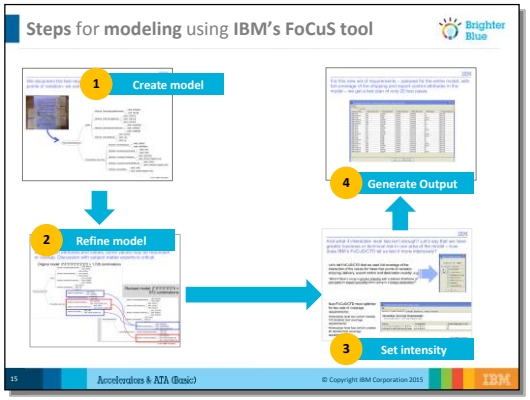
Slide Content	Instructor Guide	Use this column for your notes
Slide 12  	**Purpose:** To explain Combinatorial Test Design **Approximate Duration:** 3 minutes **What to cover:** ▪ Define the concept of CTD ▪ List the benefits of CTD ▪ Now explain the benefits achieved by using CTD using the diagram.	

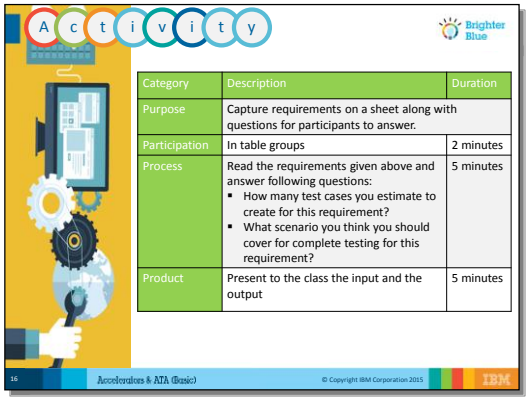
Slide Content	Instructor Guide	Use this column for your notes
Slide 13  CTD is a primary concept of modeling solution in the next generation accelerators. This slide provides an illustration to explain the concept of CTD. Some of the important aspects of CTD or modeling are: •**Point of variability:** When we get a requirement initially from the clients or business source, such as Claims Settlement so one can deduce if someone would like to create test cases and come up with millions and trillions of potential combinations because in claims settlement, there are thousands on	**Purpose:** To define CTD or modeling and explain with an example **Approximate Duration:** 8 minutes **What to cover:** ▪ Explain the concept of CTD or modeling. ▪ Explain the use of modeling with an illustration of an insurance project provided on the slide. ▪ Make sure to cover the following important aspects of CTD that are detailed in the notes section of this slide (also in the Slide Content Column of this document): ○ Point of variability ○ Interaction coverage ○ Rules \ restriction	

Slide Content	Instructor Guide	Use this column for your notes
contracts for a policy. Similarly, there are different combinations of application properties. For example, there can be 68,000 benefit requirements, and so on. Consider a scenario of manual testing in a large account, the existing tester has created around manual test cases as they initially estimated for 2000 test cases with the demand for several resources for 5-6 months and there was a quality gap as well. •Interaction coverage •Rules \ restriction CTD is an innovative approach, which yield greater business benefits. It helps reduces risk through increased coverage and improves quality. It also minimizes the number of test cases, which helps in reduction of cost, time, and resources. Based on the research after doing lots of work on several applications of similar type in large engagements, it has been estimated that CTD helps reduce the **projected test case development** and **execution costs** by 30%. This helps in reduction of 30% in the schedule, and also we have come up with 25% of increment in the test coverage.	▪ Now explain the benefits achieved by using CTD in this case. Please mention that the percentages are ballparks.	

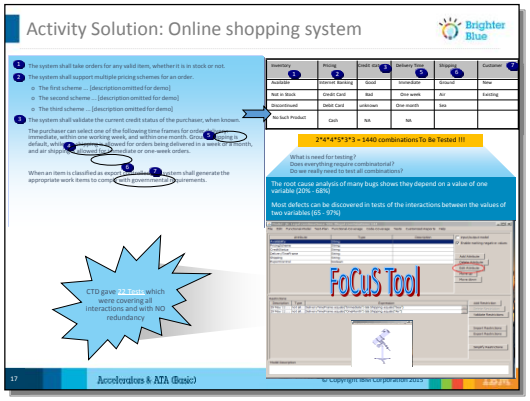
Slide Content	Instructor Guide	Use this column for your notes
Slide 14  There are different types of models used in CTD. Let us have a hi ▪ Create or identify a lean subset of test to cover a similar functionality. ▪ Create a lean solution recommendation to gain higher quality with less test. ▪ Used for maintenance purpose and enhance existing model to alter or cover more. ▪ Add more to existing model to cover gap.	**Purpose:** To list the different types of CTD models **Approximate Duration:** 2 minutes **What to cover:** ▪ Share a high-level glimpse of the different types of CTD models with the participants. ▪ Include the definitions of each type provided in the notes section of the slide (also in the Slide Content Column of this document)	

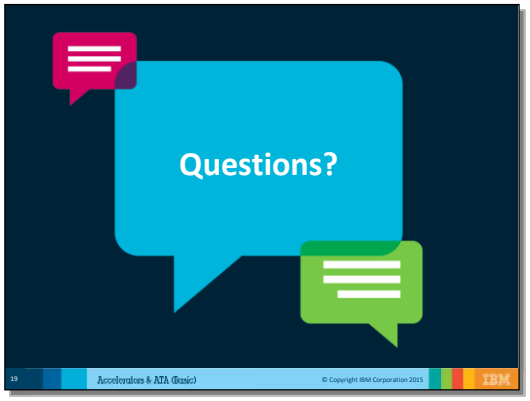
Slide Content	Instructor Guide	Use this column for your notes
Test Data Model: Create Test Data Model for setting up optimized data.		
Slide 15  	**Purpose:** To describe the common flow of modeling for CTD **Approximate Duration:** 5 minutes **What to cover:** ▪ Explain the common flow for the CTD modeling process ▪ Give an example to depict the common flow ▪ Explain each step in the flow and check back with the participants for any questions before moving on to the next step	

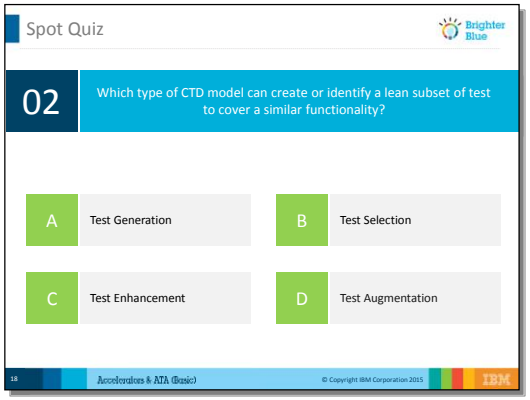
Slide Content	Instructor Guide	Use this column for your notes
Slide 16  The four steps of modeling are: 1. Create model 2. Refine model 3. Set intensity 4. Generate plan	**Purpose:** To list the steps of modeling using IBM's FoCus tool **Approximate Duration:** 2 minutes **What to cover:** ▪ Introduce the participants briefly to IBM's FoCus tool. ▪ Show them the four steps of modeling using the FoCus tool.	

Slide Content	Instructor Guide	Use this column for your notes
Slide 17  	**Activity title:** Capture requirements on a sheet along with questions for participants to answer. **Purpose:** To view a practical approach of providing estimates to the test cases **Approximate Duration:** 12-20 minutes **Additional Material or Pre-activity prep tasks**: Printed copy of requirements Rough sheets Pens **Preparation for the Activity:** Divide the class into manageable groups. **Instructions for the participants:** • Provide instructions for completing the activity.	

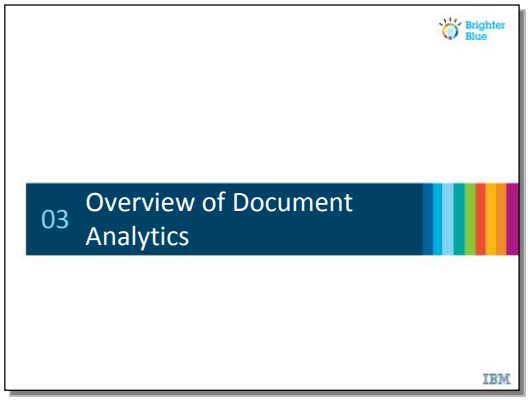
Slide Content	Instructor Guide	Use this column for your notes
	• Provide a printed requirement to the candidates and a rough sheet to complete the assignment. • Inform that each group needs to work in their table and discuss the questions presented by the instructor. • At the end of 5 minutes, each table group presents the solution to the instructor and class. **Debrief:** • Display the activity solution slide (next slide) to predict the outcome.	

Slide Content	Instructor Guide	Use this column for your notes
Slide 18  	**Purpose:** To show the solution of the Group Activity that the participants performed in the previous slide. **Approximate Duration:** 10 minutes **Debrief:** ▪ Display the activity solution slide to predict the outcome. ▪ Tell the participants that CTD gave 22 tests which covered all interactions without any redundancy. ▪ Also, point out to the participants that IBM FoCus was the tool used in this case.	

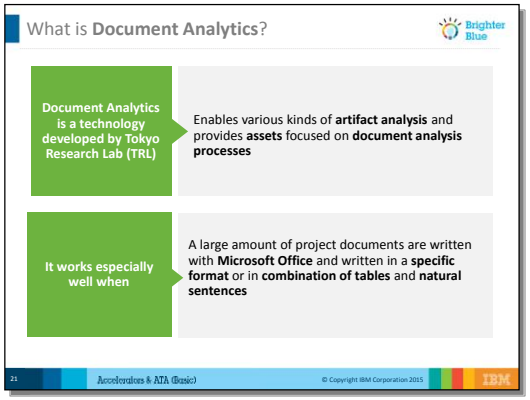
Slide Content	Instructor Guide	Use this column for your notes
Slide 19 	**Purpose:** To check knowledge of the participants for the topic covered. **Approximate Duration:** 2 minutes **What to cover:** ▪ Ask the participants the question on the slide. Treat the question as a poll - ask them to raise their hands based on their answer. ▪ The correct answer is **B. Test Selection** ▪ Explain the answer using the following explanation: Test Create or identify a lean subset of test to cover a similar functionality.	

Slide Content	Instructor Guide	Use this column for your notes
Slide 20  	**Approximate Duration:** 2 minutes **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items ▪ Use this activity to recap the key takeaways from this module	

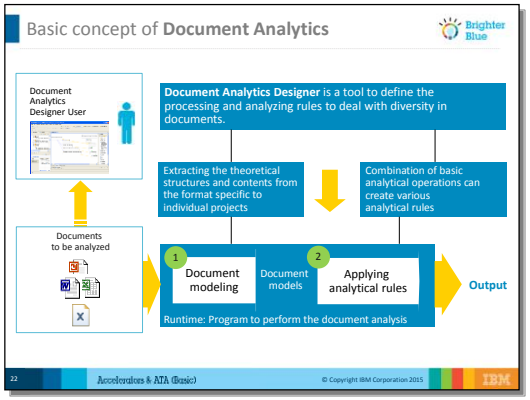
Module 03: Overview of Document Analytics

Slide Content	Instructor Guide	Use this column for your notes
Slide 21  At the end of this module, you should be able to: ▪ Recognize the features of the Document Analytics tool ▪ List the advantages of the Document Analytics tool ▪ Identify the steps performed by the Document Analytics tool to analyze documents	**Purpose:** To go over the module objectives. **Approximate Duration:** 2 minutes **What to cover:** ▪ Provide the participants with the list of topics covered in this module, which includes: ○ Recognize the features of the Document Analytics tool ○ List the advantages of the Document Analytics tool ○ Identify the steps performed by the Document Analytics tool to analyze documents	

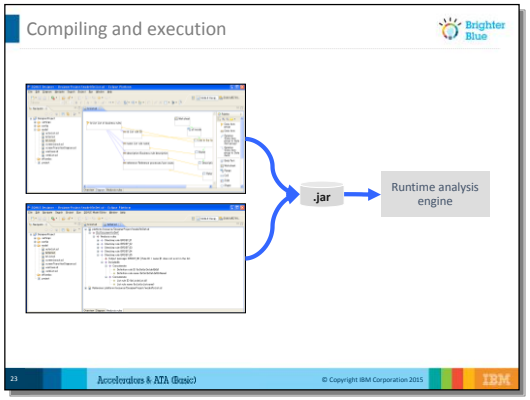
Slide Content	Instructor Guide	Use this column for your notes
▪ Identify the usage and examples of Document Analytics	○ Identify the usage and examples of Document Analytics	

Slide Content	Instructor Guide	Use this column for your notes
Slide 22  Document Analytics technology is a solution, developed by to enable analysis of diverse project artifacts (documents) applying advanced document modeling and text analysis. In particular, it provides assets focused on document analysis processes that lack analytical technologies so that various kinds of documents (requests, design and test documents during early phases of the project life cycle) can be effectively processed.	**Purpose:** To introduce the Document Analytics tool **Approximate Duration:** 5 minutes **What to cover:** ▪ Tell the participants “what is Document Analytics”. ▪ Tell them that it is tool develop by the Tokyo Research Labs (TRL) and how it helps to enable analysis of diverse project artifacts (documents) applying advanced document modeling and text analysis. ▪ Point out the scenarios when Document Analytics would work best. ▪ Include details provided in the notes section of the slide (also in the Slide Content Column of this document)	

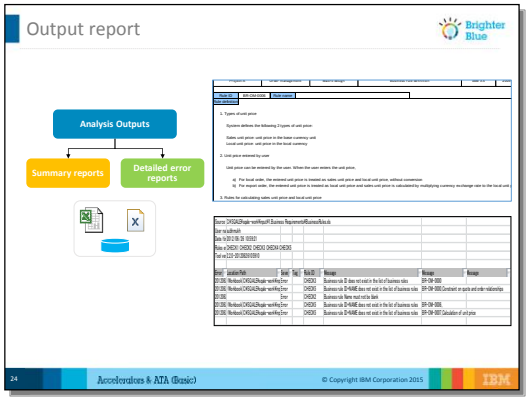
Slide Content	Instructor Guide	Use this column for your notes
It works especially well when the following situations exist: ▪ A large amount of project documents are written with Microsoft Office™. ▪ Documents are written in a specific format. ▪ Documents are written in combination of various tables and natural sentences. ▪ When analysis must be performed over multiple documents and/or various kinds of deliverables. Artifacts: All constructs created during the life cycle process of a project, including documents and code. These constructs include not only deliverables, such as products delivered to customers, but also intermediates and information necessary for process control.		

Slide Content	Instructor Guide	Use this column for your notes
Slide 23  Document Analytics helps detect simple but time-consuming defects if handled manually, and allows you to concentrate on the more complicated defects. **Document Analytics Designer** is an Eclipse workbench that provides a visual modeling tool to build document models as well as creating analytical rules. The user who works on the **Document Analytics Designer** should define the rules on data extractions, analyses and conversions on GUI depending on individual requests. Note that you do	**Purpose:** To explain how the Document Analytics tool works **Approximate Duration:** 10 minutes **What to cover:** ▪ Introduce the Document Analytics Designer to the participants. Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document). ▪ Now, explain the two key steps in the overall process flow of Document Analytics. Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document)	

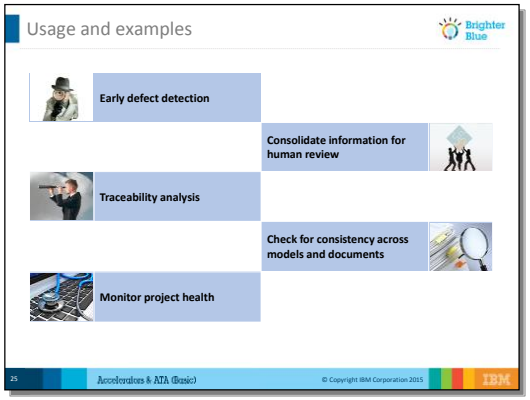
Slide Content	Instructor Guide	Use this column for your notes
not need any programming for Document Analytics Designer. **Document Analytics performs document analysis using the following two steps:** ▪ **Document modelling**: Document Analytics extracts the content, by mapping the structure and content of each different type of document that is in scope. The result of this is a Document model. ▪ **Application of analytical rules**: Document Analytics applies analysis rules to the extracted document models to modify or verify the models.		

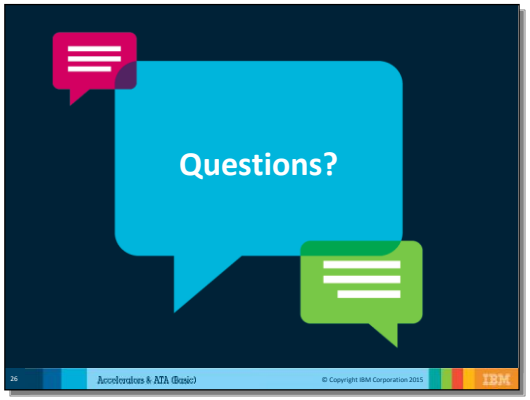
Slide Content	Instructor Guide	Use this column for your notes
Slide 24  Once the document modeling is complete and corresponding analytical rules are applied, we now run the Document Analytics using compiling and execution. This creates a Java code that is deployed to the Runtime analysis engine also known as Document Analytics runtime. Then, Document Analytics can be run on the a variety of artifacts on Document Analytics runtime by schedule and / or on demand, which then creates	**Purpose:** To cover compiling and execution **Approximate Duration:** 20 minutes **What to cover:** ▪ Explain how you will perform compiling and execution once you have performed the document modeling and defined the rules. ▪ Tell to the participants that you can run the Document Analytics using compiling and execution once you have applied the analytical rules. ▪ Also, explain that Document Analytics can be run on the a variety of artifacts on Document Analytics runtime by schedule and / or on demand, which then creates outputs like summary reports, detailed	

Slide Content	Instructor Guide	Use this column for your notes
outputs like summary reports, detailed error reports and newly generated artifacts.	error reports and newly generated artifacts. ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

Slide Content	Instructor Guide	Use this column for your notes
Slide 25  Document Analytics can be run on a variety of artifacts on an ongoing basis. This tool generates different types of output reports, which include: • Summary reports: These types of reports contain the various information, such as number of document files, size metrics of each file, and number of defects. • Detailed error reports: These types of reports contain the following information: severity, type	**Purpose:** To define the output report **Approximate Duration:** 10 - 15 minutes **What to cover:** ▪ Describe the different output reports generated by Document Analytics tool. ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

Slide Content	Instructor Guide	Use this column for your notes
of issue (rule ID), problem description, and links to the error locations. New artifacts generated.		

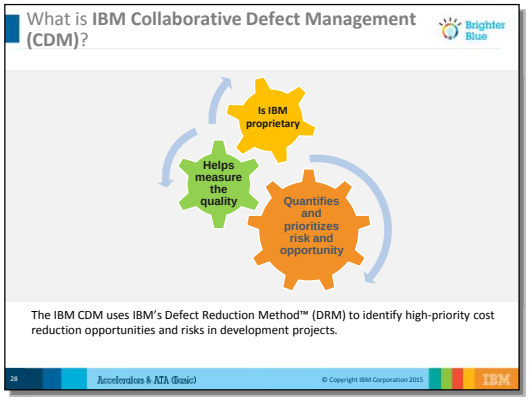
Slide Content	Instructor Guide	Use this column for your notes
Slide 26  Some of the usage and examples of Document Analytics include: **Early defect detection:** Finding errors in requirements and design documents or during test plan or test case creation phase. **:** Creating new documents using information from multiple sources : Checking relationships across development and test : Comparing business process models with specification and design documents : Measuring document quality over time	**Purpose:** To list the uses and examples of Document Analytics **Approximate Duration:** 5 minutes **What to cover:** ▪ Share with the participants the different examples of using Document Analytics ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

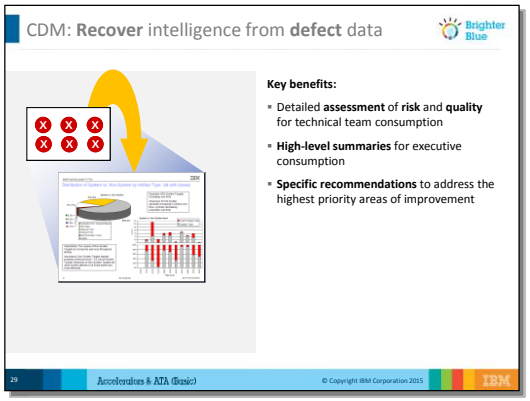
Slide Content	Instructor Guide	Use this column for your notes
Slide 27 	**Approximate Duration:** 2 minutes **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items ▪ Use this activity to recap the key takeaways from this module	

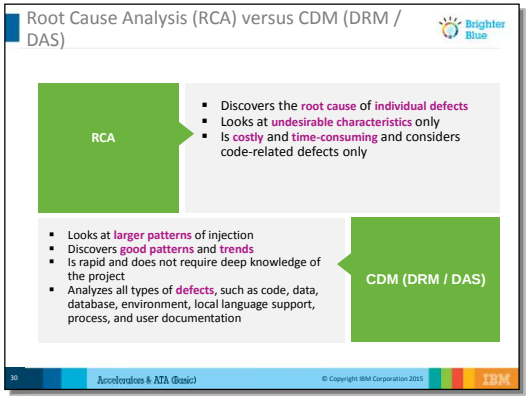
Module 04: Overview of CDM / DAS

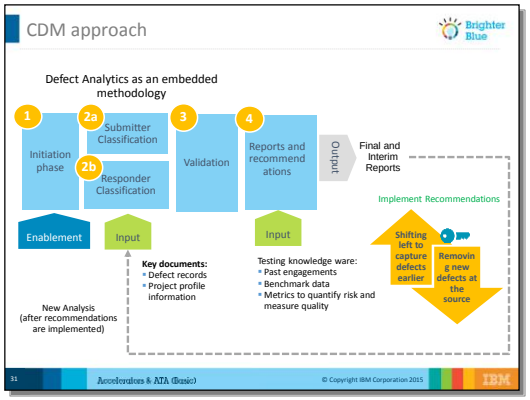
Slide Content	Instructor Guide	Use this space for your own notes
Slide 28  At the end of this module, you should be able to: ▪ Describe the features of the CDM tool ▪ Identify the benefits of using the CDM tool ▪ Differentiate between Root Cause Analysis (RCA) and CDM (DRM / DAS) ▪ Identify the phases involved in the CDM approach	**Purpose:** To go over the module objectives **Approximate Duration:** 2 minutes **What to cover:** ▪ Provide the participant with the list of topics covered in this module, which includes: ○ Describe the features of the CDM tool ○ Identify the benefits of using the CDM tool ○ Differentiate between Root Cause Analysis (RCA) and CDM (DRM / DAS) ○ Identify the phases involved in the CDM approach	

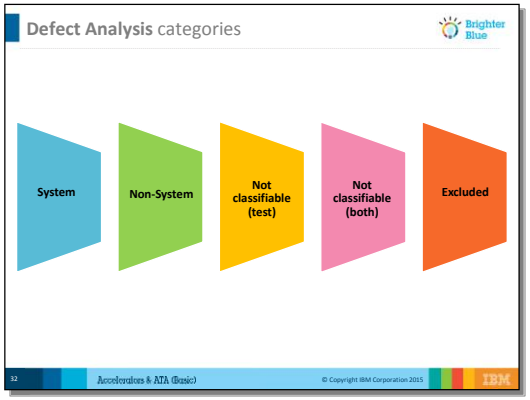
Slide Content	Instructor Guide	Use this space for your own notes
▪ Identify the different categories of defect analysis ▪ List the milestones involved in the phases of CDM	○ Identify the different categories of defect analysis ○ List the milestones involved in the phases of CDM	

Slide Content	Instructor Guide	Use this space for your own notes
Slide 29  CDM uses IBM's Defect Reduction Method™ (DRM) to identify high-priority cost reduction opportunities and risks in development projects. CDM: ▪ IBM proprietary ▪ Based on over two decades of IBM research ▪ Objectively quantifies and prioritizes risk and opportunity	**Purpose:** To define the IBM Collaborative Defect Management (CDM) tool **Approximate Duration:** 2 minutes **What to cover:** ▪ Explain the benefits of the IBM CDM tool to the participants ▪ Tell the participants that IBM CDM uses IBM's Defect Reduction Method™ (DRM) to identify high-priority cost reduction opportunities and risks in development projects. ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

Slide Content	Instructor Guide	Use this space for your own notes
▪ Objectively measures quality		
Slide 30  The IBM Collaborative Defect Management provides business value: ▪ Recommend action items that reduce risk and cost associated with development and testing ▪ Acknowledge and reinforce client strengths	**Purpose:** To list the benefits of CDM **Approximate Duration:** 5 minutes **What to cover:** ▪ State the key benefits of using CDM. ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

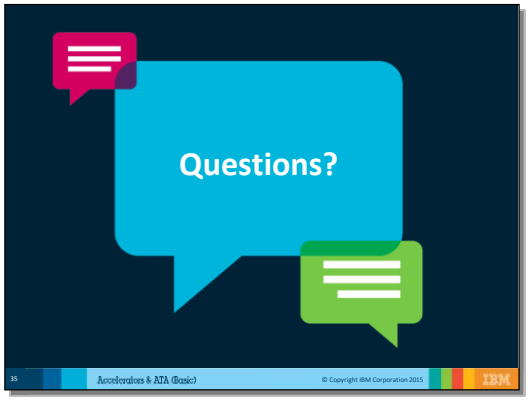
Slide Content	Instructor Guide	Use this space for your own notes
▪ Introduce clients to Testing Services offerings that increase ROI and improve quality		
Slide 31  	**Purpose:** To list the differences between Root Cause Analysis (RCA) versus CDM (DRM / DAS) **Approximate Duration:** 10 minutes **What to cover:** ▪ Compare between RCA versus CDM (DRM / DAS) ○ RCA discovers the root cause of individual defects and looks at undesirable characteristics only whereas CDM (DRM / DAS) looks at larger patterns of injection	

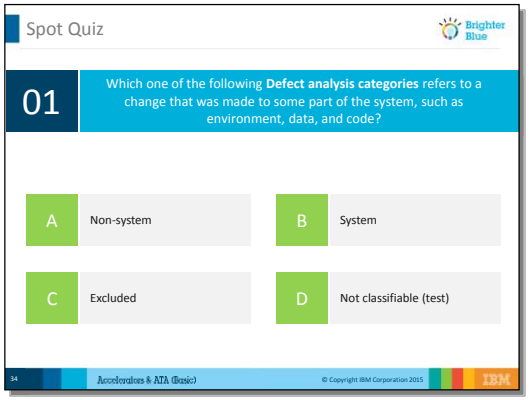
Slide Content	Instructor Guide	Use this space for your own notes
	and discovers good patterns and trends	
Slide 32  Defect Analytics is used as an embedded methodology	**Purpose:** To explain the process flow of the CDM approach **Approximate Duration:** 10 minutes **What to cover:** ▪ Describe the overall CDM approach. ▪ Explain the four steps (as outlined on the slide) in the CDM process. ▪ Explain each step in the flow and check back with the participants for any questions before moving on to the next step.	

Slide Content	Instructor Guide	Use this space for your own notes
	▪ Inform the participants that they would take a closer look at steps 2 and 3 of this process in the next slide.	
Slide 33  Some of the Defect Analysis categories are listed below: ▪ **System**: This category refers to a change was made to some part of the system,	**Purpose:** To describe the Defect Analysis categories **Approximate Duration:** 10 minutes **What to cover:** ▪ Tell the participants about the different categories of Defect Analysis ▪ Briefly explain each of these categories with an example ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

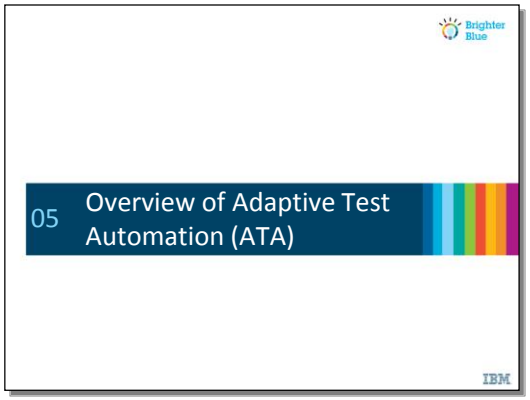
Slide Content	Instructor Guide	Use this space for your own notes
such as environment, data, code, and so on as a result of the defect ▪ **Non-System**: This category refers to a change was made to the way test was performed, either implicitly or explicitly, as a result of the defect ▪ **Not classifiable (test)**: Insufficient information was provided to classify the type of test that uncovered the defect ▪ **Not classifiable (resolution)**: Insufficient information was provided to classify the fix that was implemented ▪ **Not classifiable (both)**: Insufficient information was provided to classify the type of test that uncovered the defect and the fix that was implemented ▪ **Excluded**: The defect was not in a final (Closed) state, a final resolution was either not yet in place or not yet verified, or defect was out of scope Non-System Artifact Types: ▪ Verification Point / Expected Results ▪ Data Parameters / Inputs	▪ Name the Non-System Artifact Types provided in the notes section of the slide (also in the Slide Content Column of this document).	

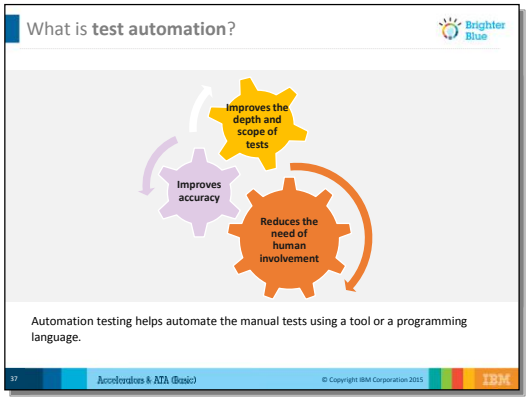
Slide Content	Instructor Guide	Use this space for your own notes
▪ Test Steps ▪ Test Case ▪ Execution Plan ▪ Resolution Plan		
Slide 34  	**Notes** **Purpose:** To define the CDM milestones **Approximate Duration:** 10 minutes **What to cover:** ▪ Explain the different milestones in the three phases of CDM to the participants. ▪ The three phases in CDM are: ○ Startup Phase ○ Perform Phase (Iterative) ○ Close Phase (Iterative)	

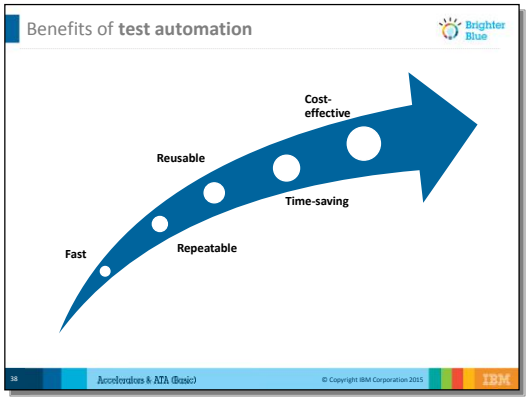
Slide Content	Instructor Guide	Use this space for your own notes
Slide 35 	**Notes** **Purpose:** To check the participants' knowledge about the topics covered. **Approximate Duration:** 2 minutes **What to cover:** • Ask the participants the question on the slide. Treat the question as a poll - ask them to raise their hands based on their answer. • The correct answer is **B. System** • Explain the answer using the following explanation: This category refers to a change was made to some part of the system, such as environment, data, code, and so on as a result of the defect.	

Slide Content	Instructor Guide	Use this space for your own notes
Slide 36  	**Notes** **Approximate Duration:** 2 minutes **What to cover:** • Ask the participants if they have any questions. • Include any questions that will be addressed later in the course as parking lot items • Use this activity to recap the key takeaways from this module	

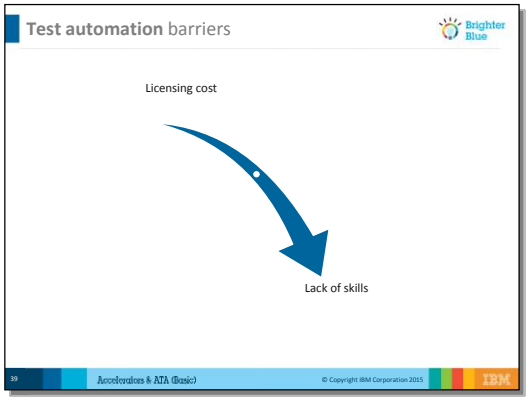
Module 05: Overview of Adaptive Test Automation (ATA)

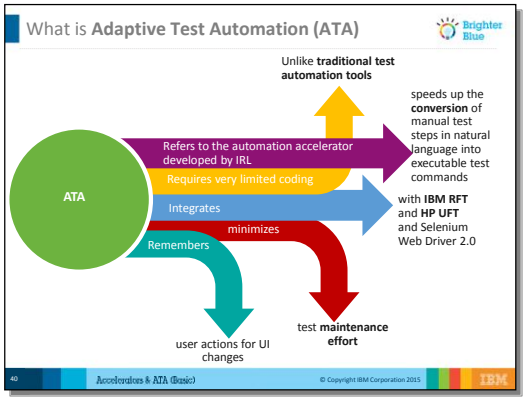
Slide Content	Instructor Guide	Use this space for your own notes
Slide 37  At the end of this module, you should be able to: ▪ Identify the benefits and barriers of test automation ▪ Recognize the features of ATA ▪ Identify the different variants of ATA ▪ Identify the features of the high-level architecture of ATA ▪ List the assumptions of implementing ATA	**Purpose:** To go over the objectives of this module **Approximate Duration:** 2 minutes **What to cover:** ▪ Provide the participant with the list of topics covered in this module, which includes: ○ Identify the benefits and barriers of test automation ○ Recognize the features of ATA ○ Identify the different variants of ATA ○ Identify the features of the high-level architecture of ATA ○ List the assumptions of implementing ATA	

Slide Content	Instructor Guide	Use this space for your own notes
Slide 38  	**Purpose:** To define test automation **Approximate Duration:** 5 minutes **What to cover:** • Briefly explain test automation to the participants. ○ Test automation improves the depth and scope of tests, improves accuracy, and reduces the need of human involvement	

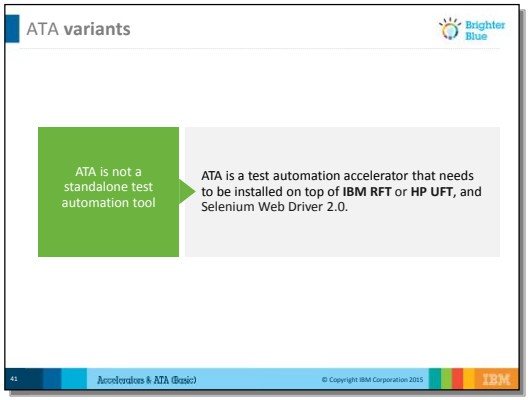
Slide Content	Instructor Guide	Use this space for your own notes
Slide 39  The benefits of test automation are listed here: **Fast**: The automated tests run fast as compared to the tests run by the humans as these kinds of tests are run by the tools. **Repeatable**: Test automation allow you to re-run the tests in a similar manner, which eliminates the risk of any human error. **Reusable**: You can reuse automated tests on different versions of the software irrespective of a change in the user interface.	**Purpose:** To cover the benefits of test automation **Approximate Duration:** 5-10 minutes **What to cover:** ▪ Briefly explain the benefits of test automation to the participants. ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

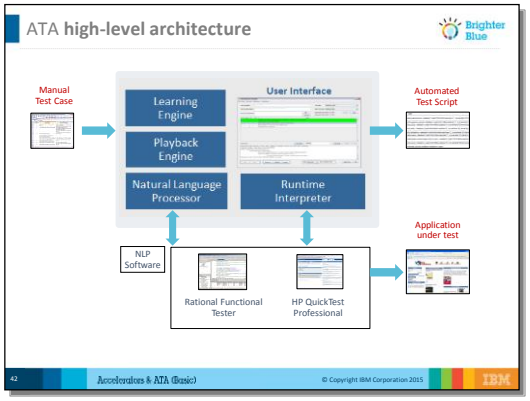
Slide Content	Instructor Guide	Use this space for your own notes
Time-saving: Test automation allows you to save time by reducing the time taken to run tests especially in case of regression testing and enable you to focus on tasks that are difficult and cannot be automated. **Cost-effective**: Automation testing also helps reduces the number of resources required for regression testing. Source: http://red-badger.com/blog/2013/02/01/benefits-of-automated-testing/		

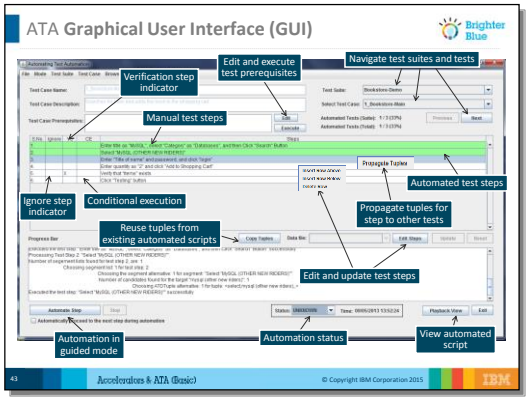
Slide Content	Instructor Guide	Use this space for your own notes
Slide 40  The key barriers of test automation are listed here: **Licensing cost:** One of the biggest barrier of automation testing is the high cost of licensing associated with the automation tool. **Lack of skills:** Automation tools are script-based, and require programming expertise. Usually, organizations do not have employees with the required skills and need to be trained or hire new employees that adds to the cost.	**Purpose:** To cover the barriers of test automation **Approximate Duration:** 5-10 minutes **What to cover:** ▪ Briefly explain the barriers of test automation to the participants. ▪ Explain that one of the biggest barrier of automation testing is the high cost of licensing associated with the automation tool. ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

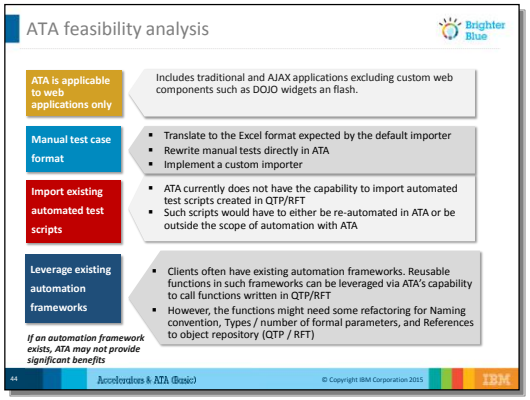
Slide Content	Instructor Guide	Use this space for your own notes
Slide 41  It is a test automation accelerator developed by IRL (IBM India Research Lab) that speeds up the conversion of manual test steps in natural language into executable test commands. Unlike traditional test automation tools, ATA requires very limited coding (coding is needed only for test steps that involve custom verification logic). ATA is fully descriptive and does not store any test object. This minimizes test maintenance effort. ATA	**Purpose:** To describe the features of Adaptive Test Automation (ATA) **Approximate Duration:** 5-10 minutes **What to cover:** ▪ Explain the following features of ATA to the participants: ○ ATA learns from “user feedback” that is it remembers user actions for UI changes and remembers these to avoid further user actions for similar changes. ○ ATA integrates with two industry standard tools—IBM RFT and HP UFT. ○ ATA is easy to learn, master and implement and saves considerable	

Slide Content	Instructor Guide	Use this space for your own notes
offers a unique feature called “Proximity-based Label Association” which ensures that during test execution ATA is able to find UI objects even after these have changed, by referring other UI objects which are associated with it. This further reduces maintenance effort. ATA learns from “user feedback” that is it remembers user actions for UI changes and remembers these to avoid further user actions for similar changes. ATA integrates with two industry standard tools—IBM RFT and HP UFT. ATA is easy to learn, master and implement and saves considerable test automation effort. ATA has already been proposed for some of the large test automation deals at IBM.	test automation effort. ATA has already been proposed for some of the large test automation deals at IBM. ○ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

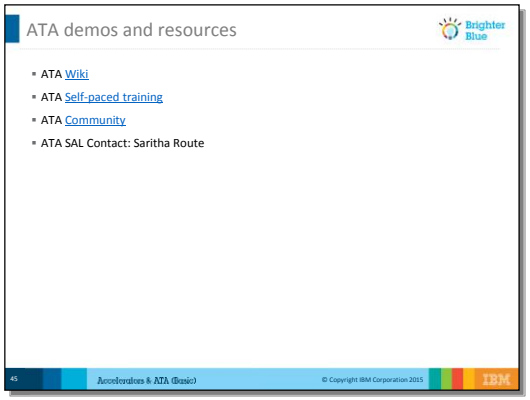
Slide Content	Instructor Guide	Use this space for your own notes
Slide 42  ATA is not a standalone test automation tool such as IBM RFT or HP UFT. It is a test automation accelerator that needs to be installed **on top** of one of these tools, and requires Selenium Web Driver 2.0 to be installed on the machine. This means that one of these tools should already been in place (installed and licensed) before ATA can be installed.	**Purpose:** To identify the different variants of ATA **Approximate Duration:** 5 minutes **What to cover:** ▪ Briefly describe the different variants for ATA to the participants. ○ It is a test automation accelerator that needs to be installed **on top** of one of these tools, and requires Selenium Web Driver 2.0 to be installed on the machine.	

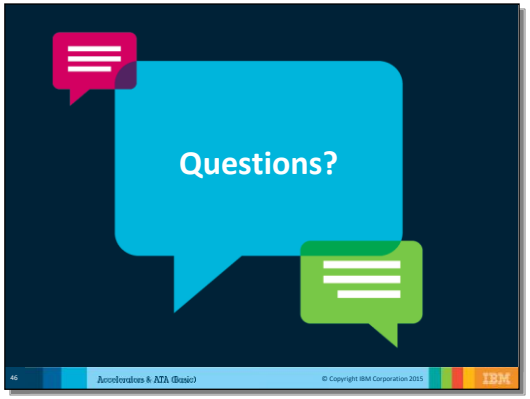
Slide Content	Instructor Guide	Use this space for your own notes
Slide 43  	**Purpose:** To cover the high-level architecture for ATA **Approximate Duration:** 8 minutes **What to cover:** ▪ Describe the high-level architecture for ATA to the participants. ▪ Explain each component of the architecture. ▪ Check with the participants if they have understood the architecture or not. Ask any one to explain it in his/her own words.	

Slide Content	Instructor Guide	Use this space for your own notes
Slide 44  	**Purpose:** To cover the GUI for ATA **Approximate Duration:** 30 minutes **What to cover:** ▪ Describe the ATA GUI to the participants using the screen capture provided on the slide. ▪ For best results, follow this up with demoing a live ATA interface on your laptop.	

Slide Content	Instructor Guide	Use this space for your own notes
Slide 45  ATA is applicable to web applications only, which includes traditional and AJAX applications excluding custom web components such as DOJO widgets an flash): ▪ ATA-QTP: Any web application that can be automated using the web add-in in QTP can also be automated using ATA-QTP ▪ ATA-RFT: Any web application that can be automated without extra plug-ins in RFT can also be automated using ATA-RFT Manual test case format:	**Purpose:** To cover the ATA feasibility analysis **Approximate Duration:** 10 - 15 minutes **What to cover:** ▪ Tell the participants about how to conduct a feasibility study of ATA. ▪ Explain each point described on the slide. ▪ Include the explanation provided in the notes section of the slide (also in the Slide Content Column of this document).	

Slide Content	Instructor Guide	Use this space for your own notes
▪ Translate to the Excel format expected by the default importer ▪ Rewrite manual tests directly in ATA ▪ Implement a custom importer Importing existing automated test scripts: ▪ ATA currently does not have the capability to import automated test scripts created in QTP/RFT ▪ Such scripts would have to either be re-automated in ATA or be outside the scope of automation with ATA Leveraging existing automation frameworks: ▪ Clients often have existing automation frameworks ▪ Reusable functions in such frameworks can be leveraged via ATA's capability to call functions written in QTP/RFT ▪ However, the functions might need some refactoring: ○ Naming convention ○ Types/number of formal parameters		

Slide Content	Instructor Guide	Use this space for your own notes
○ References to object repository (QTP/RFT) If an automation framework exists, ATA may not provide significant benefits		
Slide 46  	**Purpose:** To provide a list of reference **Approximate Duration:** 2 minutes **What to cover:** ▪ Share the list of References with the participants ○ ATA Wiki ○ ATA Self-paced training ○ ATA Community ○ ATA SAL Contact: Saritha Route ▪ Share tips on when they should use each reference / resource. ▪ Encourage them to join the relevant community and	

Slide Content	Instructor Guide	Use this space for your own notes
	leverage wiki pages and self-paced training. ▪ Point out when they should contact the SAL.	
Slide 47 	**Approximate Duration:** 2 minutes **What to cover:** ▪ Ask the participants if they have any questions. ▪ Include any questions that will be addressed later in the course as parking lot items. ▪ Use this activity to recap the key takeaways from this module and the entire track.	